PROG7311 SUMMATIVE

Technical Reflection Report

Dean Anthony Pitts
ST10449407

DevOps & Testing:

Test automation is a software testing technique that uses specialized tools and scripts to automatically execute test cases and compare actual outcomes with expected results (Bruneaux, 2026). The Global Logistics Management System makes use of CI/CD through the implementation of GitHub Actions. Automation testing is relevant to the GLMS as it incorporates three main components which all act as a possible failure point: the SQL database, the API retrieving the data from the database and the MVC front end that displays this data. These three layers have to be taken into account as any fault in these layers will cause a cascading fault amongst the rest. This is why Automation testing is essential for this implementation of CI/CD as it allows easy handling of continuous changes, a more consistent and structured approach to testing, faster testing all in an effort of preventing bugs from reaching production.

Handling of Changes

Instead of handling each small change with its own manual testing, the use of automation testing assists CI/CD pipelines by alleviating much of the manual tests of smaller updates and changes. This said, big updates still should incorporate manual testing but automated testing is perfect for the handling of countless smaller updates that if missed could bring a bug into production.

Consistent, Structured Approach

An effective CI/CD pipeline must have few variations or anomalies. Working hard to maintain process consistency ensures that expectations are met all the way downstream to the release (Verma, 2018). Having a sporadic manual testing approach can lead to some bugs ending up in production due to the testing staff being unsure of the current testing approach. Automated testing bypasses this by introducing a single structured and consistent approach made for the testing of countless smaller updates.

Faster Testing

Automated testing allows for faster and parallel testing to take place. This is due to automation testing producing results more quickly than its counterpart, manual testing. This leaves developers with more time and resources available to work on bigger features or systems.

Containerization:

Docker containers are the runnable instances created from docker images. They encapsulate the application and all its dependencies, providing a consistent and isolated runtime environment (Turner ,2026). In the context of the GLMS application, the docker containers will be split into three interconnected parts: 1st the SQL database, 2nd the backend API then the 3rd being the GLMS MVC front end which links to the API. Containerization is essential to the GLMS as it ensures consistency across development, testing and production environments by encapsulating the application, its dependencies and the runtime environment in a self-sufficient package. This ensures they are highly portable and can be used across multiple devices. This eliminates the “it works on my machine” issue as some devices may have the required dependencies and environments required for the GLMS to run while others don’t. Encapsulating the whole application along with its dependencies ensures that everyone able to access the docker file is able to run the application in the condition that the developer intended them to.

References (Reflection Report)

Bruneaux, T. (2026) *Test automation: How it affects developer productivity and code quality.* Available at: https://getdx.com/blog/test-automation/
(Accessed: 08/06/2026)

Verma, A. (2018) *Why Test Automation is Essential for CI/CD.* Available at: https://www.functionize.com/blog/why-test-automation-is-essential-for-ci-cd (Accessed: 08/06/2026)

Turner, L. (2026) What Is a Docker Container? Features and Benefits. Available at: https://www.theknowledgeacademy.com/blog/docker-container/ (Accessed: 08/06/2026)

References (GLMS)

Deschryver, T. (2026) *Tutorial: Create a controller-based web API with ASP.NET Core*. Available at: Tutorial: Create a controller-based web API with ASP.NET Core | Microsoft Learn

Umrani, A. (2024) Creating an MVC app with Firebase. Available at: Creating an MVC app with Firebase | by Aadi Umrani | Medium

Microsoft Learn. (2022) Create Data Transfer Objects (DTOs). Available at: Create Data Transfer Objects (DTOs) | Microsoft Learn

Microsoft Learn. (2026) Integration tests in ASP.NET Core. Available at: Integration tests in ASP.NET Core | Microsoft Learn

Declaration of AI Usage in my assessment:

AI Tool: Claude Sonnet 4,6 , Gemini

Purpose: Assist in scaffolding, designing and error handling parts of the GLMS

Date: 03/06/26 to 08/06/26

https://claude.ai/share/503d645e-0ce0-4796-920e-6e73ce7ccfb5
https://gemini.google.com/share/b712e62df337

Sections in Assessment	AI Tool Used	Purpose	Date Used
AuthController API	Claude	Design assistance of API AuthController allowing for calls to and from Firebase for token verification.	03/06/2026.
ContractsController API	Claude	Design assistance of API ContractsController allowing for http calls of POST, GET, PATCH, and DELETE.	03/06/2026.
LookupsController	Claude	Design assistance of API LookupsController allowing for http calls of GET.	03/06/2026.
ServiceRequestController API	Claude	Design assistance of API ServiceRequestsController allowing for http calls of POST, GET, PATCH, and DELETE.	03/06/2026.
LogisticsService API	Claude	Design assistance of API LogisticsService allowing for business logic operations and service requests handling.	03/06/2026.
Program.cs API	Claude	Design assistance and, troubleshooting of API program.cs along with all imported dependencies.	03/06/2026.
GLMS API Client MVC	Claude	Design assistance of MVC to GLMS API Client allowing for communication with front end MVC and API for db access.	03/06/2026.
Program.cs MVC	Claude	Design assistance of MVC program.cs along with troubleshooting.	04/06/2026.
API test Factory Tests	Claude	Design assistance of API Test Factory allowing for tesing of	05/06/2026.

		http calls of POST, GET, PATCH, and DELETE.	
Contracts API Test	Claude	Design assistance of ContractsAPITests allowing for testing of HTTP calls of POST, GET, PATCH, and DELETE.	05/06/2026.
File Upload Tests	Claude	Design assistance and troubleshooting of FileUploadTests allowing for testing of file upload validation in isolation.	05/06/2026.
Service Request API Test	Claude	Design assistance and troubleshooting of ServiceRequestApiTests allowing for testing of service request API endpoints.	05/06/2026.
Dockerfile API	Gemini	Design assistance of API Dockerfile along with lots of troubleshooting.	08/06/2026
Dockerfile MVC	Gemini	Design assistance of MVC Dockerfile along with lots of troubleshooting.	08/06/2026